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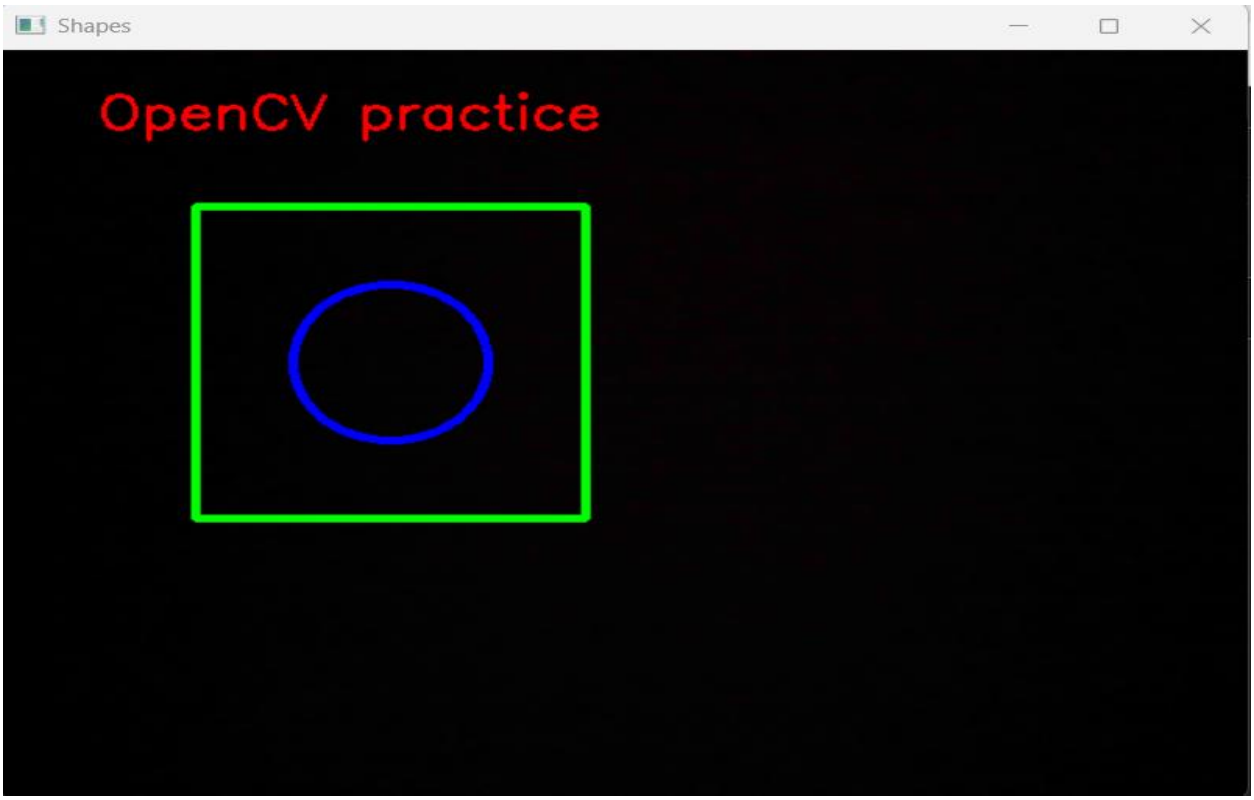
1. Webcam Video Capture

```
[1]:
import cv2

cap = cv2.VideoCapture(0, cv2.CAP_DSHOW)

print("Camera opened:", cap.isOpened())

while True:
    ret, frame = cap.read()

    if not ret:
        print("Failed to grab frame")
        break

    cv2.imshow("Webcam", frame)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

```
Camera opened: True
```

2. Drawing Shapes (Rectangle & Circle)

```
[ ]:
import cv2

cap = cv2.VideoCapture(0, cv2.CAP_DSHOW)

while True:
    ret, frame = cap.read()

    if not ret:
        break

    cv2.rectangle(frame, (100,100), (300,300), (0,255,0), 3)
    cv2.circle(frame, (200,200), 50, (255,0,0), 3)

    cv2.putText(frame, "OpenCV practice", (50,50),
                cv2.FONT_HERSHEY_SIMPLEX, 1, (0,0,255), 2)

    cv2.imshow("Shapes", frame)

    if cv2.waitKey(1) == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```



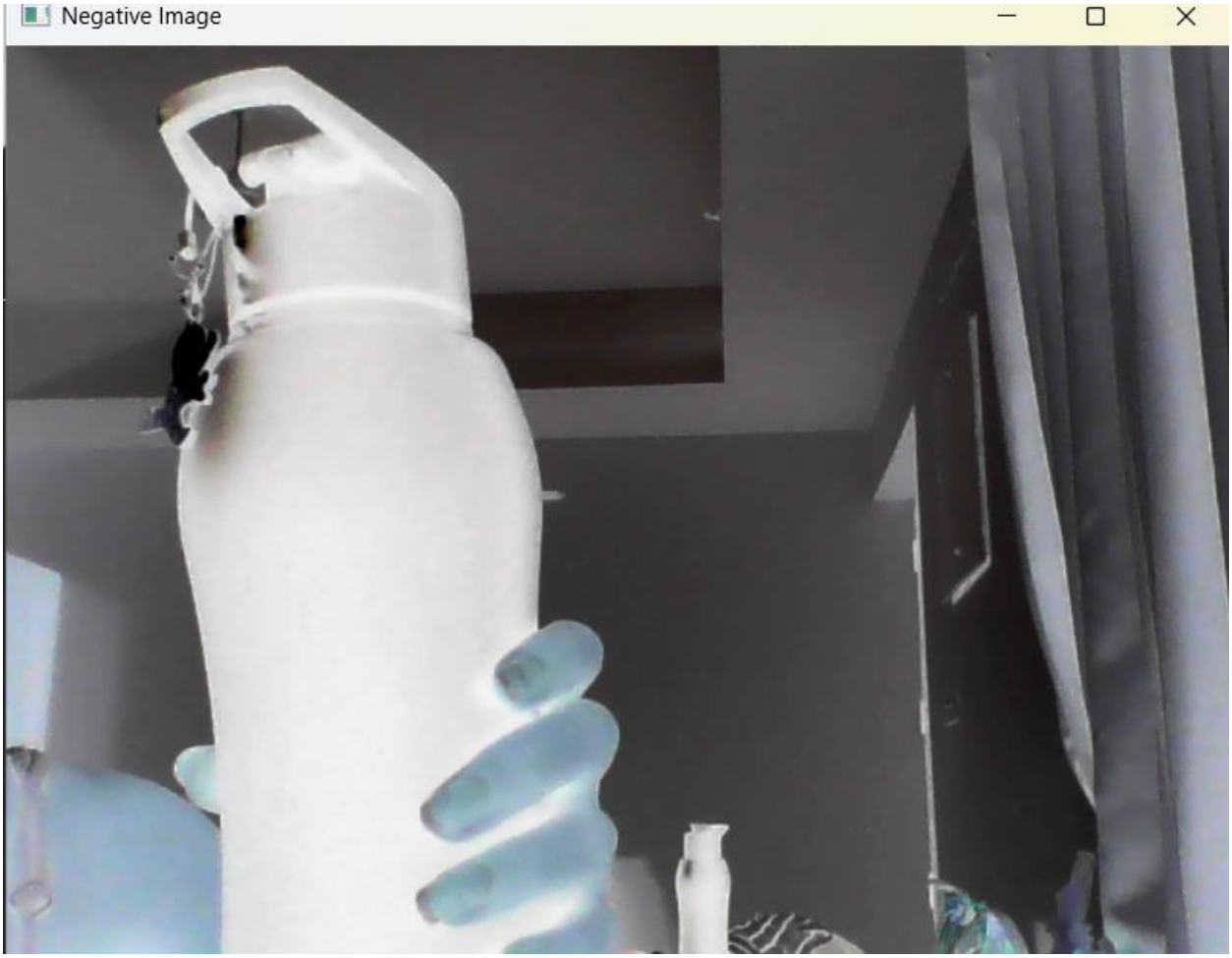
3. Canny Edge Detection

[3]:

```
import cv2

cap = cv2.VideoCapture(0, cv2.CAP_DSHOW)

while True:
    ret, frame = cap.read()

    if not ret:
        break

    edges = cv2.Canny(frame, 100, 200)

    cv2.imshow("Edges", edges)

    if cv2.waitKey(1) == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

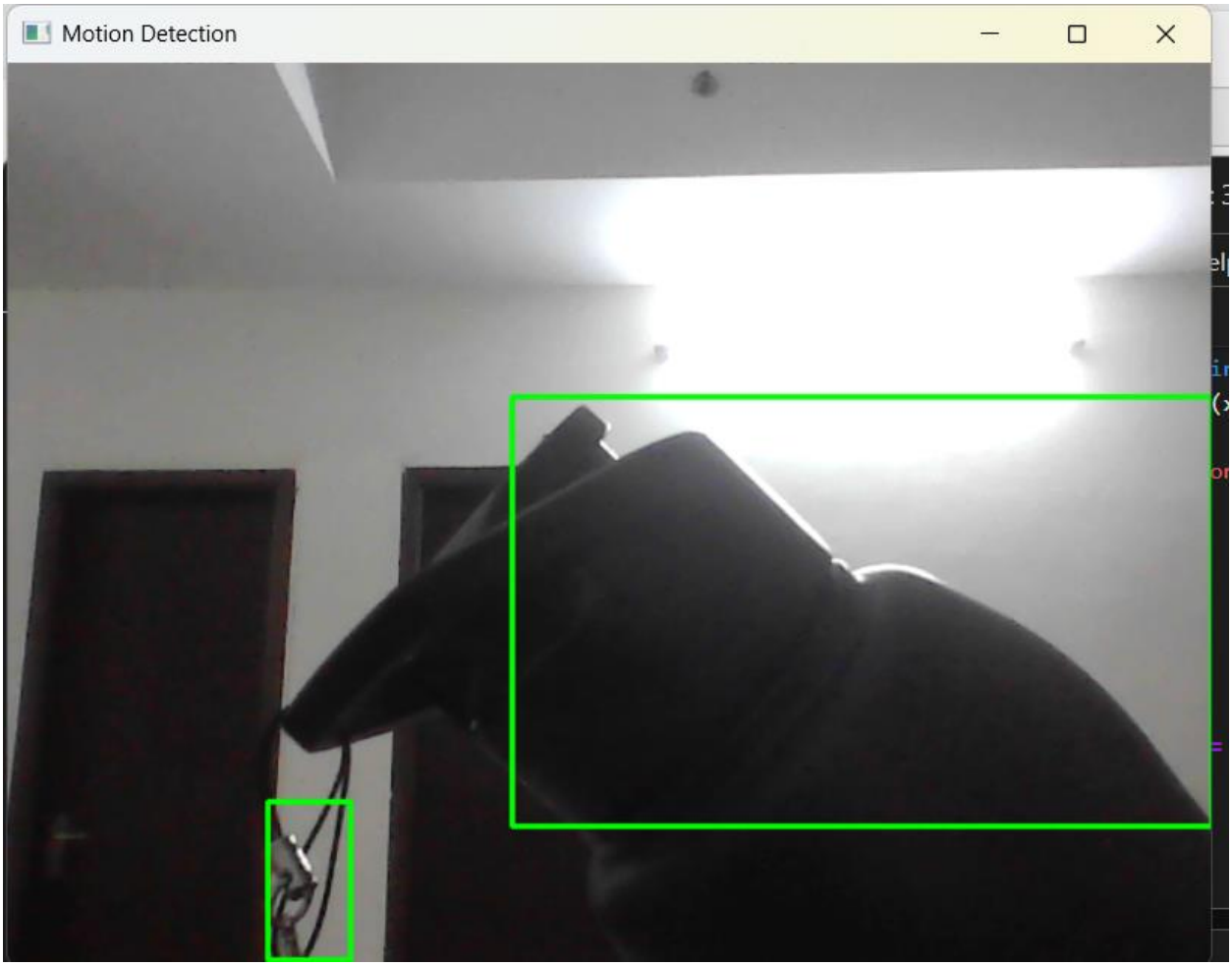
4. Negative Image Transformation

```
[5]:
import cv2

cap = cv2.VideoCapture(0, cv2.CAP_DSHOW)

while True:
    ret, frame = cap.read()

    if not ret:
        break

    negative = 255 - frame

    cv2.imshow("Original", frame)
    cv2.imshow("Negative Image", negative)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```



5. Motion Detection

```
import cv2

cap = cv2.VideoCapture(0, cv2.CAP_DSHOW)

# check if camera opened
if not cap.isOpened():
    print("Error: Camera not accessible")
    exit()

ret, frame1 = cap.read()
ret, frame2 = cap.read()

while ret:

    diff = cv2.absdiff(frame1, frame2)

    gray = cv2.cvtColor(diff, cv2.COLOR_BGR2GRAY)

    blur = cv2.GaussianBlur(gray, (5,5), 0)

    _, thresh = cv2.threshold(blur, 20, 255, cv2.THRESH_BINARY)

    dilated = cv2.dilate(thresh, None, iterations=3)

    contours, _ = cv2.findContours(dilated, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE)

    for contour in contours:

        if cv2.contourArea(contour) < 1000:
            continue

        x, y, w, h = cv2.boundingRect(contour)
        cv2.rectangle(frame1, (x,y), (x+w,y+h), (0,255,0), 2)

    cv2.imshow("Motion Detection", frame1)

    frame1 = frame2
    ret, frame2 = cap.read()

    if not ret:
        break

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```


Motion Detection